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REG NO:23MIS0333

AGILE DEVELOPMENT PROCESS AND DEVOPS LAB FAT EXAM

SET-6

PROBLEM STATEMENT:

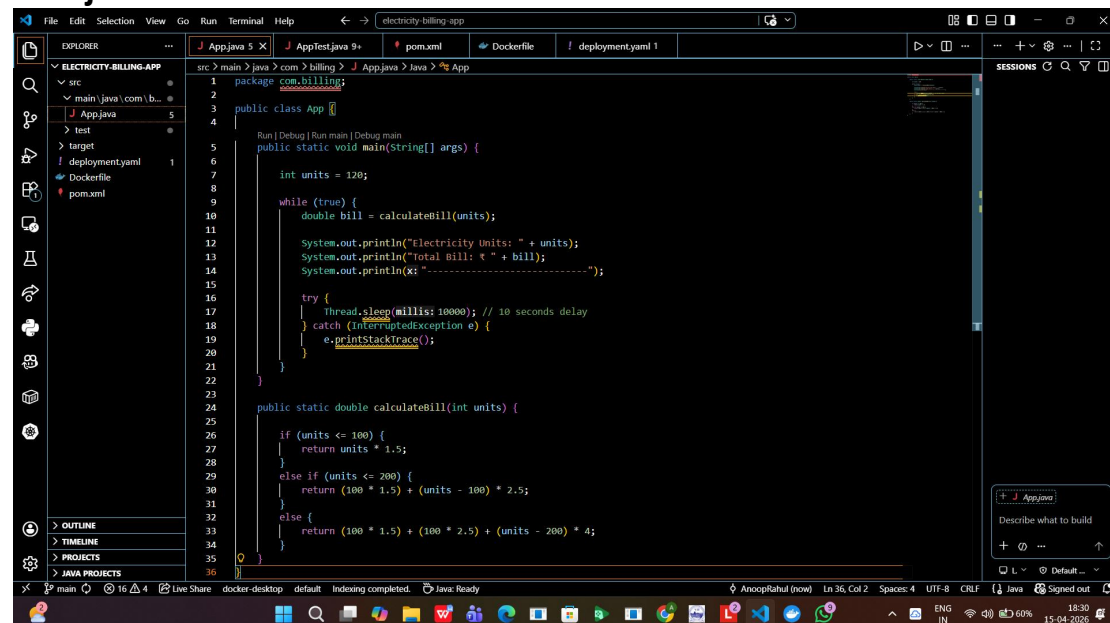
ENSURE ELECTRICITY USAGE IS CALCULATED AND BILLED CORRECTLY.

SOLUTION:

Java programming:(BILLING CALCULATION LOGIC)

1. JAVA PROGRAMMING DEVELOPMENT

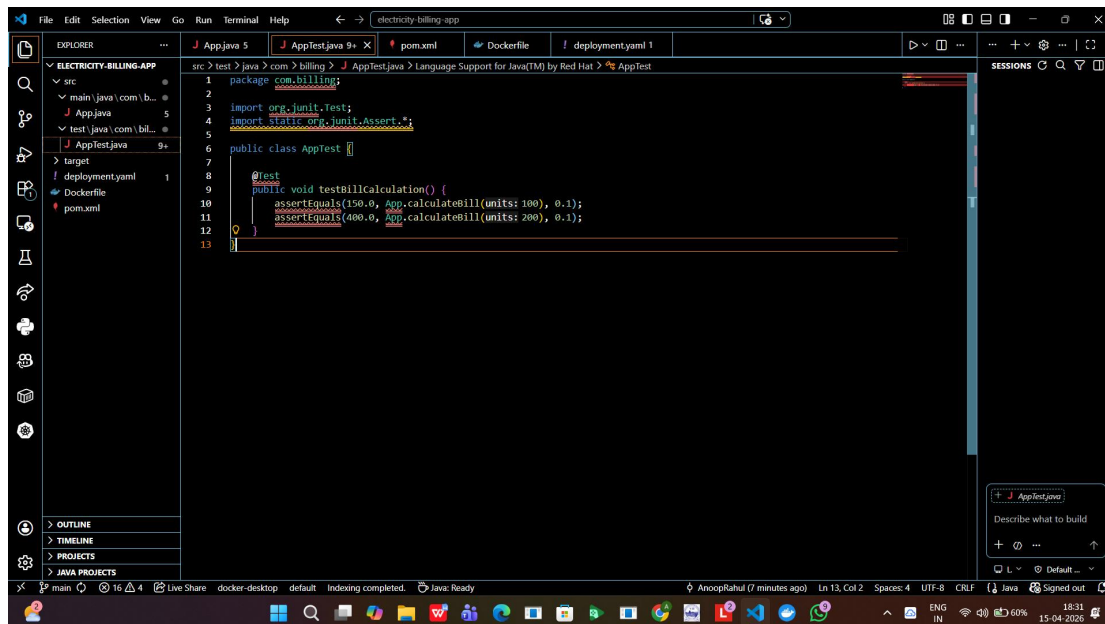
APP.java:



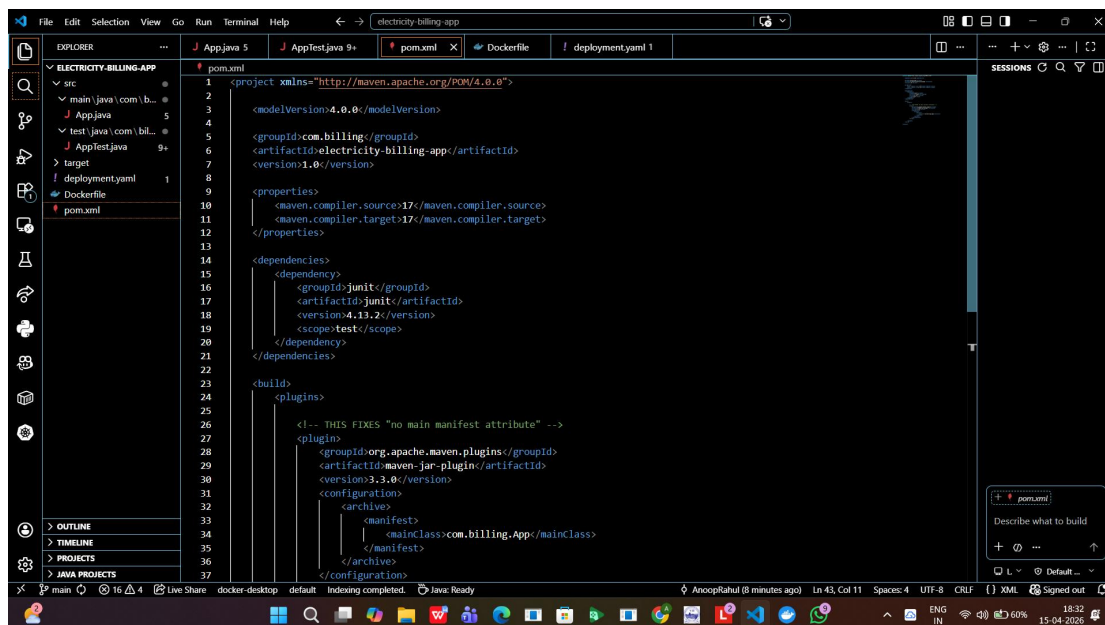
```
1 package com.billing;
2
3 public class App {
4     public static void main(String[] args) {
5         int units = 120;
6
7         while (true) {
8             double bill = calculateBill(units);
9
10            System.out.println("Electricity Units: " + units);
11            System.out.println("Total Bill: ₹ " + bill);
12            System.out.println(x: "-----");
13
14            try {
15                Thread.sleep(10000); // 10 seconds delay
16            } catch (InterruptedException e) {
17                e.printStackTrace();
18            }
19        }
20    }
21
22    public static double calculateBill(int units) {
23        if (units <= 100) {
24            return units * 1.5;
25        }
26        else if (units <= 200) {
27            return (100 * 1.5) + (units - 100) * 2.5;
28        }
29        else {
30            return (100 * 1.5) + (100 * 2.5) + (units - 200) * 4;
31        }
32    }
33 }
```

2. JUNIT TESTING:

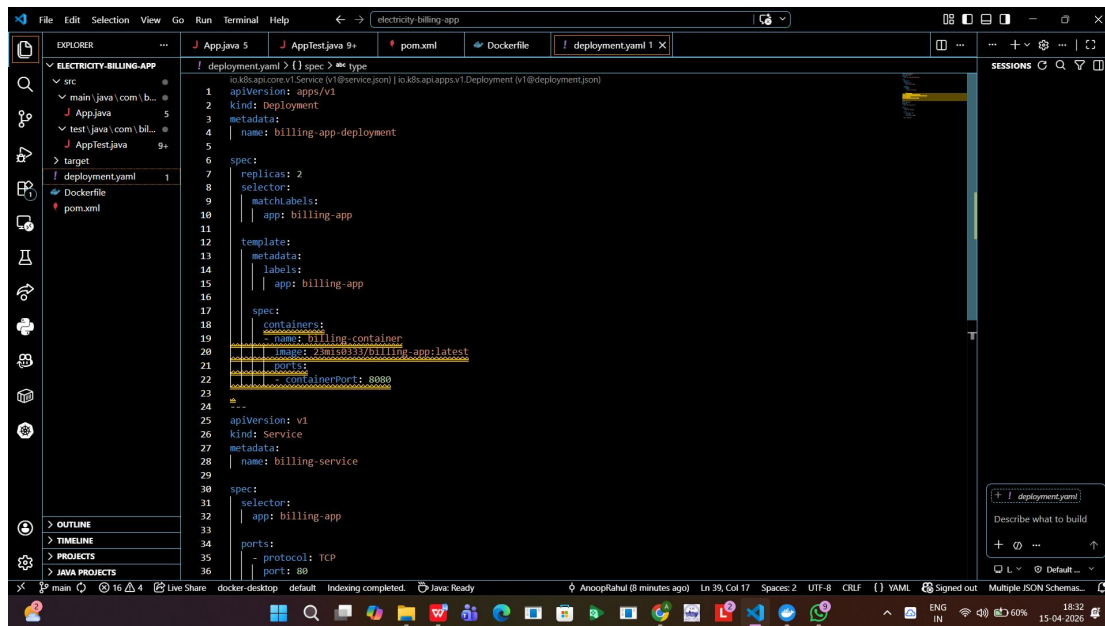
AppTest.java:



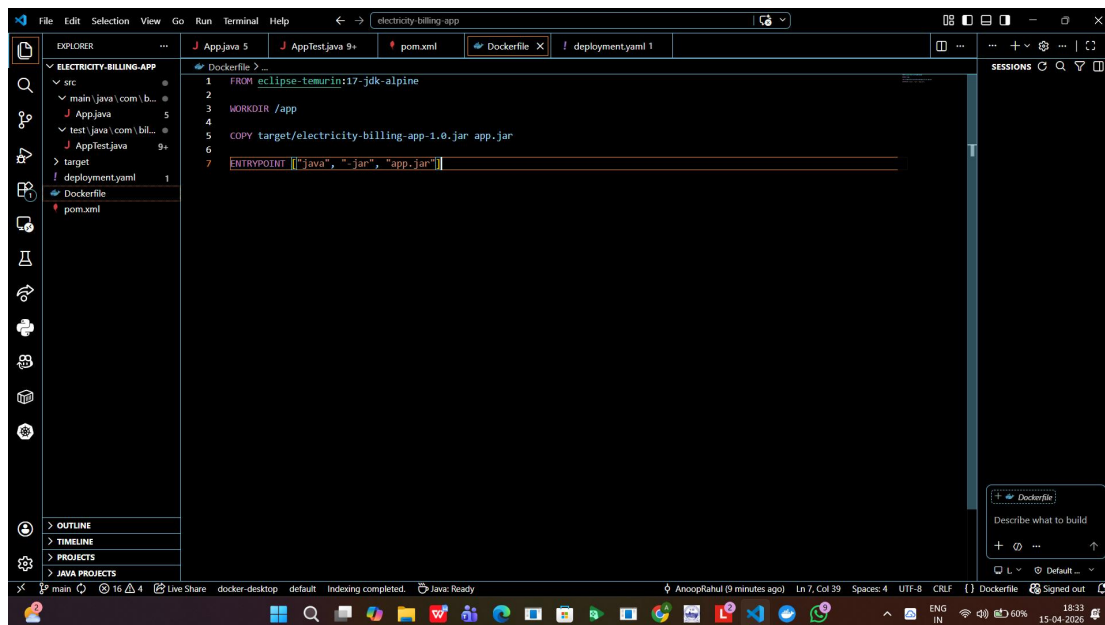
Pom.xml:



Deployment.yaml:



Dockerfile:

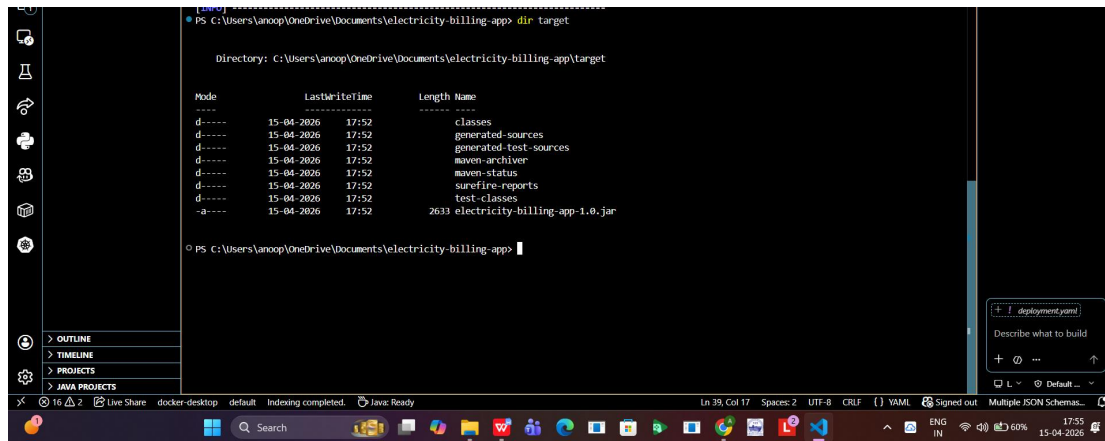


MNV clean package:

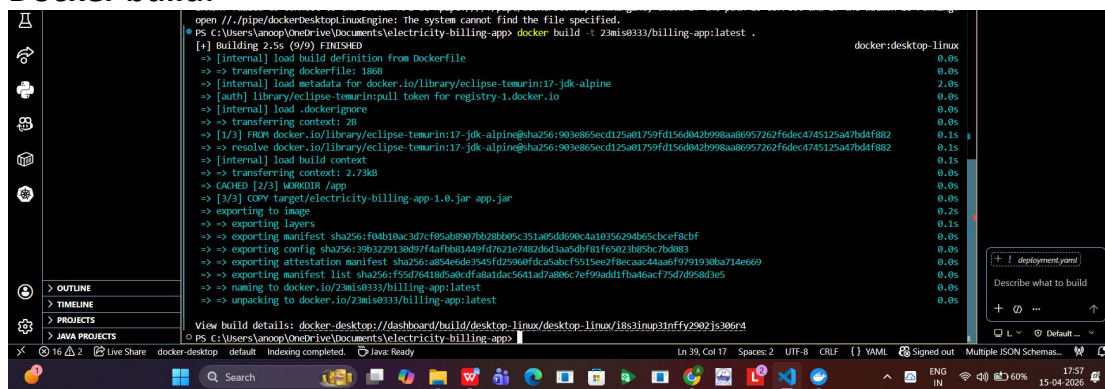
```
PS C:\Users\lanoop\OneDrive\Documents\electricity-billing-app> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----t com.billing:electricity-billing-app >-----
[INFO] Building electricity-billing-app 1.0
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ electricity-billing-app ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ electricity-billing-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\lanoop\OneDrive\Documents\electricity-billing-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ electricity-billing-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ electricity-billing-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\lanoop\OneDrive\Documents\electricity-billing-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ electricity-billing-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ electricity-billing-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.5.4/surefire-junit4-3.5.4.jar (3.3 kB at 6.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar (3.4 kB at 42.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.jar (2.7 kB at 23.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar (3.4 kB at 42.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.jar (2.7 kB at 23.0 kB/s)
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ electricity-billing-app ---
[INFO] Building jar: C:\Users\lanoop\OneDrive\Documents\electricity-billing-app\target\electricity-billing-app-1.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO] Total time: 5.902 s
[INFO] Finished at: 2026-04-15T17:52:43+05:30
[INFO]
[INFO] -----
```

```
PS C:\Users\lanoop\OneDrive\Documents\electricity-billing-app> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----t com.billing:electricity-billing-app >-----
[INFO] Building electricity-billing-app 1.0
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ electricity-billing-app ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ electricity-billing-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\lanoop\OneDrive\Documents\electricity-billing-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ electricity-billing-app ---
[INFO] Recompiling the module because of changed source code.
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[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ electricity-billing-app ---
[INFO] Recompiling the module because of changed dependency.
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[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.5.4/surefire-junit4-3.5.4.jar (3.3 kB at 6.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar (3.4 kB at 42.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.jar (2.7 kB at 23.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar (3.4 kB at 42.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.jar (2.7 kB at 23.0 kB/s)
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ electricity-billing-app ---
[INFO] Building jar: C:\Users\lanoop\OneDrive\Documents\electricity-billing-app\target\electricity-billing-app-1.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO] Total time: 5.902 s
[INFO] Finished at: 2026-04-15T17:52:43+05:30
[INFO]
[INFO] -----
```

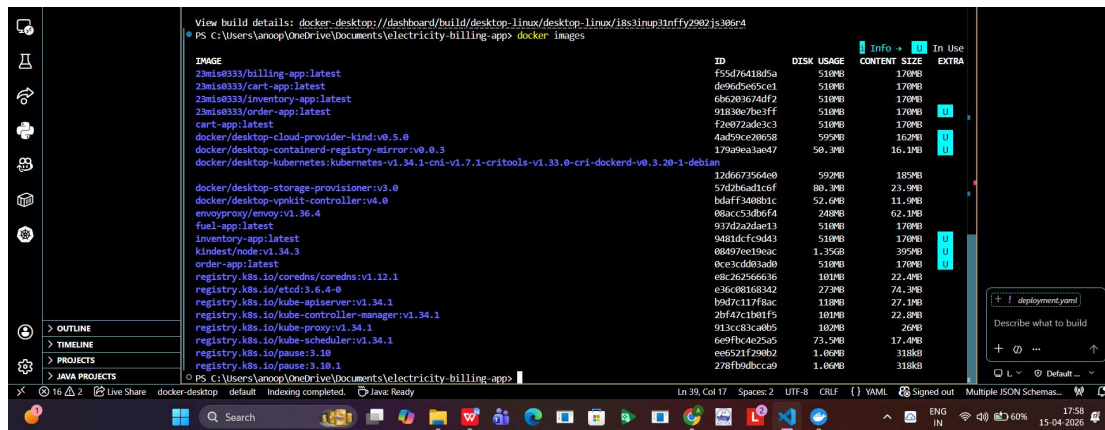
Dir target:



Docker build:



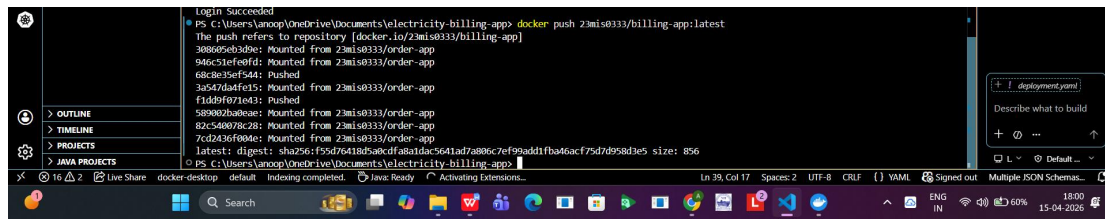
DOCKER IMAGE:



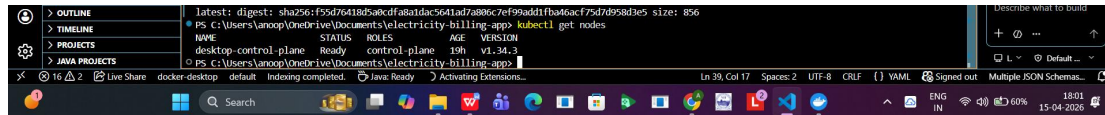
DOCKER LOGIN:



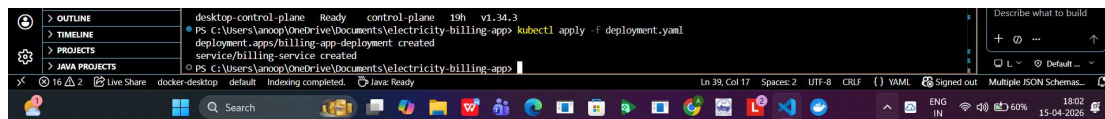
IMAGE PUSH:



Check Kubernetes cluster

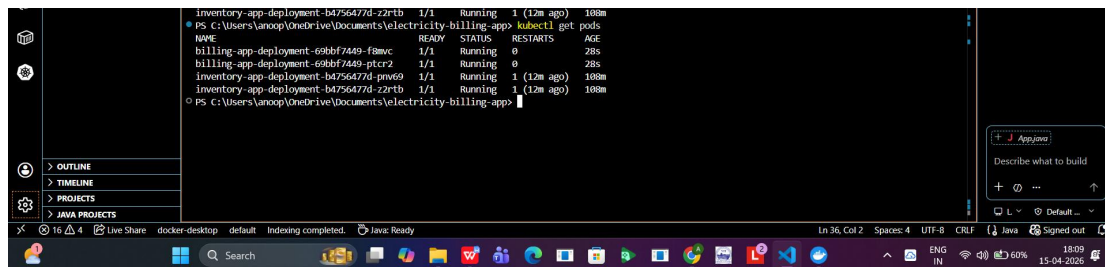


Deploy to Kubernetes:

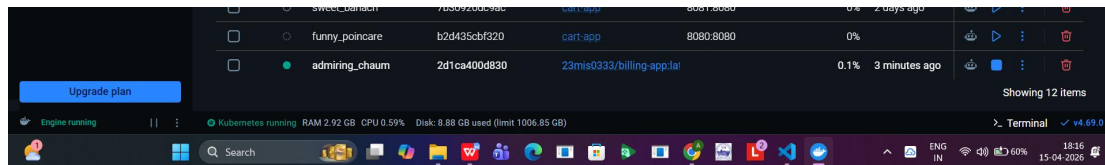


4.DEPLOY SERVICES ON DOCKER AND KUBERNETES:

Check pods:

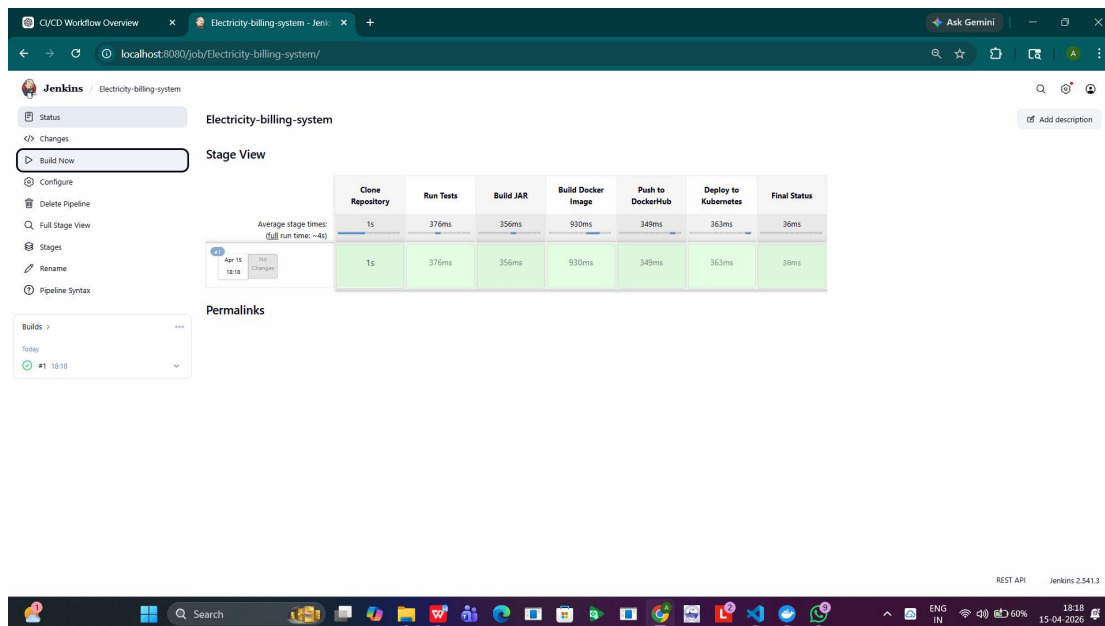


Kubectl logs:

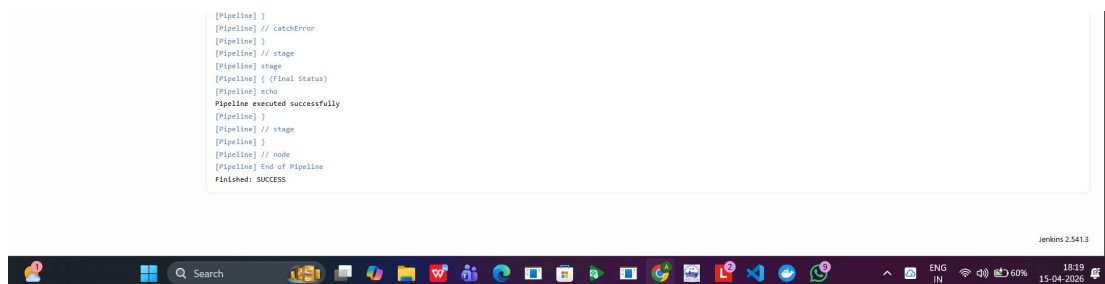


3. CI/CD PIPELINE CREATION AND EXECUTION:

Jenkins build:



Jenkins console output:



Jenkins code:

Pipeline

Advanced

Definition

Pipeline script

```
Script ?
1 pipeline {
2   agent any
3
4   tools {
5     maven 'Raven'
6   }
7
8   environment {
9     DOCKER_IMAGE = '2mls0333/billing-app:latest'
10  }
11
12  stages {
13
14    stage('Clone Code') {
15      steps {
```

☒ Use Groovy Sandbox

[Pipeline Syntax](#)

Advanced

Advanced

Save Apply

Pipeline

Advanced

Definition

Pipeline script

```
Script ?
53 }
54
55 stage('Verify Container Output') {
56   steps {
57     script {
58       bat 'docker rm -f temp-billing || ver >nul'
59       bat 'docker run -d --name temp-billing 2mls0333/billing-app:latest'
60
61       powershell 'Start-Sleep -Seconds 10'
62
63       bat 'docker logs temp-billing | findstr "Total Bill"'
64     }
65   }
66 }
67
```

☒ Use Groovy Sandbox

[Pipeline Syntax](#)

Advanced

Advanced

Save Apply

Triggers

Pipeline

Advanced

```
Script ?
77 }
78 }
79 }
80
81 post {
82   success {
83     echo 'Pipeline SUCCESS ✓'
84   }
85   failure {
86     bat 'docker stop temp-billing || ver >nul'
87     bat 'docker rm temp-billing || ver >nul'
88     echo 'Pipeline FAILED ✗'
89   }
90 }
91
```

☒ Use Groovy Sandbox

[Pipeline Syntax](#)

Advanced

Advanced

Save Apply

REST API Jenkins 2.541.3

Github pushing:

```
PS C:\Users\anoop\OneDrive\Documents\electricity-billing-app> git init
Initialized empty Git repository in C:\Users\anoop\OneDrive\Documents\electricity-billing-app\git\
PS C:\Users\anoop\OneDrive\Documents\electricity-billing-app> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-compile/createdFiles.lst', LF will be replaced by CRLF th
e next time git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by
CRLF the next time git touches it
warning: in the working copy of 'target/surefire-reports/TEST-com.billing.AppTest.xml', LF will be replaced by CRLF the next time git touches it
PS C:\Users\anoop\OneDrive\Documents\electricity-billing-app> git commit -m "Initial commit - Electricity Billing System"
(master root-commit) 90b2125 Initial commit - Electricity Billing System
15 files changed, 211 insertions(+)
create mode 100644 Dockerfile
create mode 100644 deployment.yaml
create mode 100644 pom.xml
create mode 100644 src/main/java/com/billing/App.java
create mode 100644 src/test/java/com/billing/AppTest.java
create mode 100644 target/classes/com/billing/App.class
create mode 100644 target/electricity-billing-app-1.0.jar
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.billing.AppTest.xml
```

```
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.billing.AppTest.xml
create mode 100644 target/test-classes/com/billing/AppTest.class
PS C:\Users\anoop\OneDrive\Documents\electricity-billing-app> git remote add origin https://github.com/AnoopRahul/electricity-billing-app.git
Enumerating objects: 41, done.
Counting objects: 100% (41/41), done.
Delta compression using up to 12 threads
Compressing objects: 100% (21/21), done.
Writing objects: 100% (41/41), 8.37 KiB | 1.05 MiB/s, done.
Total 41 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AnoopRahul/electricity-billing-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\anoop\OneDrive\Documents\electricity-billing-app>
```

Github repo:

